CNN-based Neural Receivers for OFDM-SIMO systems over CDL channel models

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***Abstract*— Neural networks are universal function approximators, but whether they can replace or improve conventional model-based PHY receivers remains an open practical question.Here we study a neural receiver for OFDM-SIMO system over CDL channel models. The performance and robustness of the receivers are examined through a series of simulations. The main conclusion is that the neural receiver is a strong learned alternative to practical channel-estimation-based receiver, while it remains sensitive to some channel/domain shifts.**

I. System Model

The transmitter structure is identical across the three receiver configurations. Bits are first encoded with a NR-compliant LDPC encoder and mapped onto QPSK symbols. The resulting data symbols are then arranged on an OFDM resource grid and transmitted over a CDL channel model. The neural receiver is compared with two conventional systems, LMMSE equalizer with perfect CSI and least-squares channel estimation from pilot symbols. For the conventional receivers, the equalized symbols from are demapped to yield LLRs for the LDPC decoder. One transmit antenna and two receiver antennas are used to simplify the structure of the neural receiver.

II. CNN-based Neural Receiver

A. Structure

The neural receiver is a convolutional residual network that operates directly on the received OFDM resource grid. The complex samples from the two receive antennas are split into real and imaginary components, and the noise variance is appended as an additional input feature, forming a five-channel tensor. This tensor is first mapped to 128 feature channels by a (3×3) convolution, processed by four residual convolutional blocks with LayerNorm and ReLU activations, and finally projected by another (3×3) convolution to bit-wise LLR estimates. The output contains two LLR grids, one for each coded bit per resource element.

B. Training

The neural receiver is trained end-to-end over randomly sampled SNR values. At each iteration, random coded bits are mapped to QPSK symbols, placed on the OFDM resource grid, transmitted through the CDL-OFDM channel, and processed by the neural receiver to produce bit-wise LLRs. The training metric is the negative bit-metric decoding rate, computed from the binary cross-entropy between the transmitted coded bits and predicted LLRs. Model parameters are updated using Adam with a learning rate of (10^{-3}).

III. Results

Sionna was used to generate the comparisons below [1].

A. Comparisons with traditional receivers

First, the BLER of three receivers is examined. It can be observed that the neural receiver outperformed the LS-based LMMSE receiver by approximately 3.5 dB at 10^{-2} BLER. Comparing with the LMMSE with perfect CSI, the performance of the neural receiver is comparable to that of the conventional receiver in the low-SNR regime, while it is outperformed in the high SNR. In the low-SNR regime, the LMMSE equalizer, which fails to capture the full noise pattern, and the demapper, which relies on deterministic constellation, suffers noise enhancement and imperfect estimation. In contrast, the neural receiver effectively learns to compensate the noise and estimation errors. However, in the high-SNR regime, the approximation-based neural receivers cannot match the theory-driven LMMSE equalizers. Nevertheless, well-designed CNN-based receivers, such as the DeepRX with deeper layers, can still match the LMMSE equalizer, even at high-SNR regime [2].

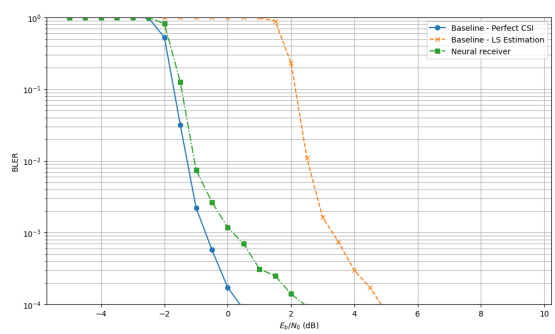

Figure 1: Coded BLER performance of three configurations.

B. Robustness under channel model mismatch

Under channel model mismatch, the neural receiver shows mixed generalization behavior. It maintains comparable performance on CDL-A, CDL-D, and CDL-E, although it is trained on CDL-C. In CDL-A, the performance is slightly better than on the training channel. However, CDL-B shows a clear degradation, requiring 3.5 dB higher SNR to reach the same BLER target of 10^{-2}. This suggests that the receiver is sensitive to channel statistics that differ substantially from the training distribution. The reason might be unveiled when one examine the cumulative distribution fucntion of the realizations of the channel models. Reliable deployment would require training over a mixture of CDL models or applying domain adaptation for poorly matched environments as demonstrated in the next section.

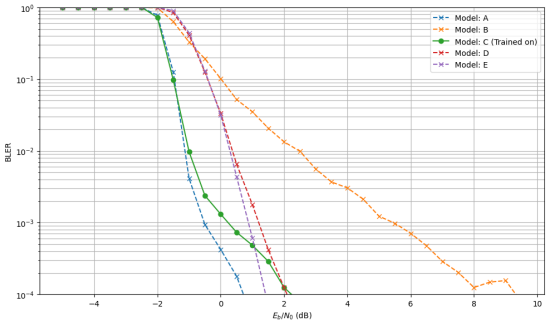

Figure 2: Performance of the trained RX on five CDL channel models.

C. Domain adaptation

Fine-tuning on CDL-B provides a clear BLER improvement over the pretrained receiver. Before adaptation, the CDL-B model gives a BLER of approximately 9.9×10^{-3} at 2.5 dB. The BLER gradually decreases as training progresses, reaching values around 2.5×10^{-3} to 4.0×10^{-3} after 1000 steps. This corresponds to roughly a threefold reduction in block error rate. The positive BLER gain after about 350 steps indicates that the pretrained receiver can be effectively adapted to a mismatched channel with a small number of additional updates. These results support fine-tuning as a strategy for specific channel conditions.

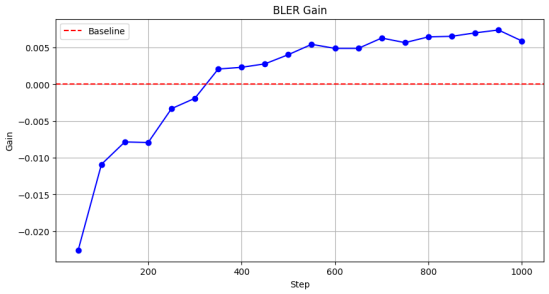

Figure 3: BLER gain when fine-tuning the model on CDL-B.

IV. Conclusion

In this report, we examined the performance of a neural receiver. Compared with the LMMSE receiver using LS channel estimation, it achieves a clear BLER improvement and reduces the SNR required to reach a given error-rate target. The results also show that the receiver can generalize to several unseen CDL channel models, although performance degrades under stronger channel mismatch, particularly for CDL-B and larger delay spreads. Fine-tuning on the mismatched CDL-B channel significantly improves BLER, demonstrating the potential of transfer learning for adapting neural receivers to new propagation environments.

Future work should investigate training over randomized mixtures of CDL models, delay spreads, mobility values, and antenna configurations to improve robustness. Ablation should also be conducted to quantify the contribution of each architectural component. In addition, pruning and quantization should be explored to reduce model size and computational cost. Further studies could also compare complexity, latency, and memory footprint against conventional receivers to assess practical deployment feasibility.

References

[1] J. Hoydis, S. Cammerer, F. Ait Aoudia, M. Nimier-David, L. Maggi, G. Marcus, A. Vem, and A. Keller, "Sionna," 2022. https://nvlabs.github.io/sionna/.

[2] M. Honkala, D. Korpi, and J. M. J. Huttunen, "Deeprx: Fully convolutional deep learning receiver," *IEEE Transactions on Wireless Communications*, vol. 20, no. 6, pp. 3925–3940, 2021.